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How to Install Apache Spark on Windows



By [Goran Jevtic](#) Published: October 10, 2024

Topics: [Big Data](#), [Windows](#)

Apache Spark is an **open-source** framework for processing large volumes of batch and streaming data from multiple sources. It is used in distributed computing for **machine learning**, **data analytics**, and graph processing tasks.

Learn how to install Apache Spark on Windows and verify the installation works.



Prerequisites

- A system running Windows 10 or 11.
- A user account with administrator privileges.
- Access to Windows **Command Prompt** or PowerShell.
- A tool to **extract .tar files**, such as 7-Zip or WinRAR.

Install and Set Up Apache Spark on Windows

To set up Apache Spark, you must install [Java](#), download the Spark package, and set up environment variables. Python is also required to use Spark's Python API called PySpark.

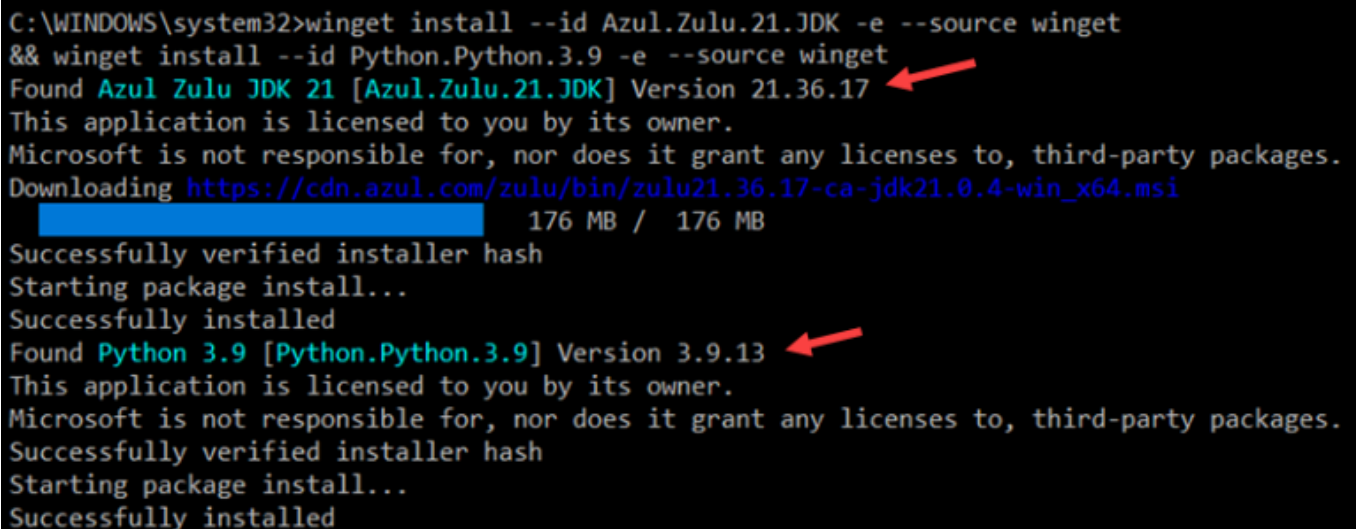
If you already have Java 8 (or later) and [Python 3](#) (or later) installed, you can skip the first step of this guide.

Step 1: Install Spark Dependencies

Using the Windows [winget](#) utility is a convenient way to install the necessary dependencies for Apache Spark:

1. Open [Command Prompt or PowerShell](#) as an Administrator.
2. Enter the following command to install the [Azul Zulu OpenJDK 21](#) (Java Development Kit) and **Python 3.9**:

```
winget install --id Azul.Zulu.21.JDK -e --source winget && winget install --id Python.Python.3.9 -e --source winget
```



```
C:\WINDOWS\system32>winget install --id Azul.Zulu.21.JDK -e --source winget
&& winget install --id Python.Python.3.9 -e --source winget
Found Azul Zulu JDK 21 [Azul.Zulu.21.JDK] Version 21.36.17
This application is licensed to you by its owner.
Microsoft is not responsible for, nor does it grant any licenses to, third-party packages.
Downloading https://cdn.azul.com/zulu/bin/zulu21.36.17-ca-jdk21.0.4-win_x64.msi
176 MB / 176 MB
Successfully verified installer hash
Starting package install...
Successfully installed
Found Python 3.9 [Python.Python.3.9] Version 3.9.13
This application is licensed to you by its owner.
Microsoft is not responsible for, nor does it grant any licenses to, third-party packages.
Successfully verified installer hash
Starting package install...
Successfully installed
```

Adjust the command if you need a specific Java or [Python version](#) or a different OpenJDK vendor.

3. The OpenJDK installation is in the `C:\Program Files\Zulu\zulu-21` **folder** by default. The space in the path can cause issues when launching Apache Spark. Avoid this by moving the installation to a folder without spaces. Use the following command to create a new *Zulu* folder in the [root directory](#) and move the installation:

```
mkdir C:\Zulu && robocopy "C:\Program Files\Zulu\zulu-21" "C:\Zulu\zulu-21" /E /MOVE
```

4. Verify the installation by checking the Java and Python versions:

```
java -version && python --version
```

```
C:\WINDOWS\system32>java -version && python --version
openjdk version "21.0.4" 2024-07-16 LTS
OpenJDK Runtime Environment Zulu21.36+17-CA (build 21.0.4+7-LTS)
OpenJDK 64-Bit Server VM Zulu21.36+17-CA (build 21.0.4+7-LTS, mixed mode, sharing)
Python 3.9.13
```

The output confirms your system is using **OpenJDK 21** and **Python 3.9**.

Step 2: Download Apache Spark

To download the latest Apache Spark release:

1. Open a browser and navigate to the official [Apache Spark download page](#).
2. The latest Spark version is selected by default. At the time of writing, the latest version is **Spark 3.5.3 for Hadoop 3.3**.
3. Click the **spark-3.5.3-bin-hadoop3.tgz** download link.

The screenshot shows the Apache Spark download page. The Apache Spark logo is at the top left. Navigation links include Download, Libraries, Documentation, Examples, Community, Developers, and GitHub. The main heading is "Download Apache Spark™". Below it, there are four numbered steps: 1. Choose a Spark release (3.5.3 (Sep 24 2024) is selected), 2. Choose a package type (Pre-built for Apache Hadoop 3.3 and later is selected), 3. Download Spark: spark-3.5.3-bin-hadoop3.tgz (indicated by a red arrow and a red circle with the number 3), and 4. Verify this release using the 3.5.3 signatures, checks, and project release KEYS by following these procedures. A note states: "Note that Spark 3 is pre-built with Scala 2.12 in general and Spark 3.2+ provides additional pre-built distribution with Scala 2.13." Below this, there is a section "Link with Spark" with the text: "Spark artifacts are hosted in Maven Central. You can add a Maven dependency with the following coordinates:" followed by a code block:

```
groupId: org.apache.spark
artifactId: spark-core_2.12
version: 3.5.3
```

 At the bottom, there is a section "Installing with PyPi" with the text: "PySpark is now available in pypi. To install just run pip install pyspark."

4. Select a location from a list of mirror servers to begin the download.



We suggest the following location for your download:

<https://dlcdn.apache.org/spark/spark-3.5.3/spark-3.5.3-bin-hadoop3.tgz>

Alternate download locations are suggested below.

It is essential that you [verify the integrity](#) of the downloaded file using the PGP signature (`.asc` file) or a hash (`.md5` or `.sha*` file).

HTTP

<https://dlcdn.apache.org/spark/spark-3.5.3/spark-3.5.3-bin-hadoop3.tgz>

BACKUP SITES

<https://dlcdn.apache.org/spark/spark-3.5.3/spark-3.5.3-bin-hadoop3.tgz>

5. Save the file, for example, to the *Downloads* folder.

Step 3: Verify Spark Software File

To verify that you are working with an unaltered, uncorrupted file, check its **checksum**:

1. Navigate to the Spark Download page and open the **checksums** link, preferably in a new tab.

2. Open Command Prompt and use the `cd` command to navigate to the folder where you downloaded Apache Spark. For example, if the file is in the *Downloads* folder, enter:

```
cd C:\Users\username\Downloads
```

Replace `username` with your actual Windows username.

3. Use the following command to calculate the downloaded file's checksum:

```
certutil -hashfile spark-3.5.3-bin-hadoop3.tgz SHA512
```

```
C:\Users\<username>\Downloads>certutil -hashfile spark-3.5.3-bin-hadoop3.tgz SHA512
SHA512 hash of spark-3.5.3-bin-hadoop3.tgz:
3b041e538282c15ad3b0ab2f05d558e1be2d16e047c2a98ca9c08176c4a50e7101bd2199aec97a66fdabf4b3f49
6948ee6d
CertUtil: -hashfile command completed successfully.
```

The system displays a long alphanumeric code followed by the message `Certutil: -hashfile completed successfully`.

4. Manually compare the checksum output with the one on the Apache Spark website.

If they match, the file is legitimate.

Step 4: Install Apache Spark

To install Apache Spark, extract the downloaded file to a desired location:

1. For example, create a new *Spark* folder in the root of the *C:* drive using the following command:

```
cd \&& mkdir Spark
```

2. Use 7-Zip to extract the Spark *.tar* file from the *.tgz* compressed file and place it in the *Spark* folder:

```
"C:\Program Files\7-Zip\7z.exe" x "C:\Users\username\Downloads\spark-3.5.3-bin-hadoop3.tgz" -o
C:\Spark
```

```
C:\>"C:\Program Files\7-Zip\7z.exe" x "C:\Users\<username>\Downloads\spark-3.5.3-bin-hadoop3.tgz" -oC:\Spark
7-Zip 22.00 (x64) : Copyright (c) 1999-2022 Igor Pavlov : 2022-06-15
Scanning the drive for archives:
1 file, 400864419 bytes (383 MiB)
Extracting archive: C:\Users\<username>\Downloads\spark-3.5.3-bin-hadoop3.tgz
--
Path = C:\Users\<username>\Downloads\spark-3.5.3-bin-hadoop3.tgz
Type = gzip
Headers Size = 10
Everything is Ok
Size:      445972480
Compressed: 400864419
```

Replace `username` in the file path with your username.



Note: The file paths are wrapped in double quotes because the username and folder names contain spaces.

3. Extract the Spark **binaries** from the Spark **.tar** file:

```
"C:\Program Files\7-Zip\7z.exe" x "C:\Spark\spark-3.5.3-bin-hadoop3.tar" -oC:\Spark
```

```
C:\>"C:\Program Files\7-Zip\7z.exe" x "C:\Spark\spark-3.5.3-bin-hadoop3.tar" -oC:\Spark
7-Zip 22.00 (x64) : Copyright (c) 1999-2022 Igor Pavlov : 2022-06-15

Scanning the drive for archives:
1 file, 445972480 bytes (426 MiB)

Extracting archive: C:\Spark\spark-3.5.3-bin-hadoop3.tar
--
Path = C:\Spark\spark-3.5.3-bin-hadoop3.tar
Type = tar
Physical Size = 445972480
Headers Size = 1328128
Code Page = UTF-8
Characteristics = GNU LongName ASCII

Everything is Ok

Folders: 196
Files: 1825
Size:      444177357
Compressed: 445972480
```

4. Use the `dir` command to list the contents of the *Spark* folder:

```
cd C:\Spark && dir
```

```
C:\>cd C:\Spark && dir
Volume in drive C has no label.
Volume Serial Number is 323E-E686

Directory of C:\Spark

10/08/2024  04:11 PM    <DIR>          .
10/08/2024  04:11 PM    <DIR>          ..
09/09/2024  07:33 AM    <DIR>          spark-3.5.3-bin-hadoop3
10/08/2024  02:24 PM             445,972,480 spark-3.5.3-bin-hadoop3.tar
               1 File(s)      445,972,480 bytes
               3 Dir(s)  31,055,310,848 bytes free
```

The `spark-3.5.3-bin-hadoop3` folder contains the necessary files to run Spark.

Step 5: Add winutils.exe File

The **winutils** utility enables Apache Spark and other **Hadoop-based** tools to run on Windows. You need to download the *winutils.exe* file that matches the Hadoop version used by your Spark installation:

1. Create a *hadoop\bin* folder in the *C:* drive to store the *winutils.exe* file:

```
cd \ && mkdir C:\hadoop\bin
```

2. Use the `curl` command to download the file from the **winutils GitHub repository** into the newly created folder:

```
curl --ssl-no-revoke -L -o C:\hadoop\bin\winutils.exe https://github.com/cdarlint/winutils/raw/master/hadoop-3.3.5/bin/winutils.exe
```

```
C:\>curl --ssl-no-revoke -L -o C:\hadoop\bin\winutils.exe
https://github.com/cdarlint/winutils/raw/master/hadoop-3.3.5/bin/winutils.exe
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           % Dload  % Upload   Total   Spent    Left     Speed
  0     0    0     0    0     0      0     0 --:--:-- --:--:-- --:--:--    0
100 110k 100 110k    0     0 90063      0  0:00:01  0:00:01 --:--:-- 156k
```

In this example, the Hadoop version is **3.3.5**. If necessary, edit the command to correspond to the Hadoop version your Spark installation is using.

3. Verify that the file has been downloaded:

```
cd C:\hadoop\bin && dir
```

```
C:\WINDOWS\system32>cd C:\hadoop\bin && dir
Volume in drive C has no label.
Volume Serial Number is 323E-E686

Directory of C:\hadoop\bin

10/08/2024  03:44 PM    <DIR>          .
10/08/2024  03:44 PM    <DIR>          ..
10/08/2024  03:44 PM             112,640 winutils.exe
               1 File(s)             112,640 bytes
               2 Dir(s)  30,871,511,040 bytes free
```

The *winutils.exe* file is listed in the folder.

Step 6: Configure Environment Variables

Add the Spark, Java, and Hadoop locations to your system's `Path` environment variable to run the Spark shell directly from the CLI.

You can **edit environment variables** manually using the Windows UI. However, setting User Variables from Command Prompt can be faster and more efficient.

Use the `setx` command to set the `SPARK_HOME` User Variable:

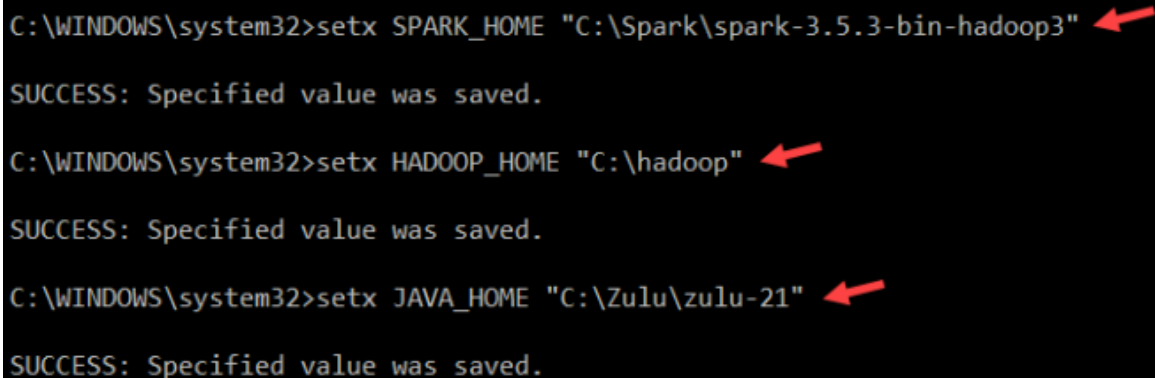
```
setx SPARK_HOME "C:\Spark\spark-3.5.5-bin-hadoop3"
```

To set `HADOOP_HOME`, enter:

```
setx HADOOP_HOME "C:\hadoop"
```

To set `JAVA_HOME`, type:

```
setx JAVA_HOME "C:\Zulu\zulu-21"
```



```
C:\WINDOWS\system32>setx SPARK_HOME "C:\Spark\spark-3.5.3-bin-hadoop3"
SUCCESS: Specified value was saved.

C:\WINDOWS\system32>setx HADOOP_HOME "C:\hadoop"
SUCCESS: Specified value was saved.

C:\WINDOWS\system32>setx JAVA_HOME "C:\Zulu\zulu-21"
SUCCESS: Specified value was saved.
```

The paths in the commands are set based on the previous installation steps. If you used different versions and file paths, adjust the commands accordingly.

Enter the following command to retrieve the current Path from the registry and store it in the `oldPath` variable:

```
for /f "tokens=2*" %A in ('reg query "HKCU\Environment" /v Path') do set "oldPath=%B"
```

Append the Spark, Hadoop, and Java paths to the `Path` variable without overwriting existing entries:

```
setx Path "%oldPath%;%SPARK_HOME%\bin;%HADOOP_HOME%\bin;%JAVA_HOME%\bin"
```



```
C:\WINDOWS\system32>for /f "tokens=2*" %A in ('reg query "HKCU\Environment" /v Path') do set "oldPath=%B"

C:\WINDOWS\system32>set "oldPath=C:\Program Files\Zulu\zulu-21\bin\;C:\Program Files\Common Files\Oracle\Java
:Program Files (x86)\Common Files\Oracle\Java\javapath;c:\Program Files (x86)\NVIDIA Corporation\PhysX\Com
ot%\system32;%SystemRoot%;%SystemRoot%\System32\Wbem;%SYSTEMROOT%\System32\WindowsPowerShell\v1.0\;%SYSTEMR
\OpenSSH\;C:\Program Files (x86)\Calibre2\;C:\Program Files\Intel\WiFi\bin\;C:\Program Files\Common Files\I
Common\;C:\Program Files\Microsoft Web Platform Installer\;C:\Python27\;C:\Cassandra\apache-cassandra-3.11.6
\php;C:\ProgramData\ComposerSetup\bin;%NVM_HOME%;%NVM_SYMLINK%;C:\Program Files\nodejs\;C:\Program Files\Po
in;C:\Program Files (x86)\Bitwise SSH Client;C:\Program Files\PuTTY\;C:\Program Files (x86)\NetSarang\Xshel
\
\AppData\Local\Programs\Python\Python39;C:\Users\
ripts;"

C:\WINDOWS\system32>setx Path "%oldPath%;%SPARK_HOME%\bin;%HADOOP_HOME%\bin;%JAVA_HOME%\bin"

SUCCESS: Specified value was saved.
```

Restart Command Prompt to apply the changes. In rare instances, a full system reboot may be required.

Step 7: Launch Spark

To start the Spark Scala shell, enter the following:

spark-shell

This opens an interactive shell interface with Spark notifications and information. The output includes details about the Spark version, configuration, and available resources.

```
C:\WINDOWS\system32>spark-shell  
Setting default log level to "WARN".  
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).  
24/10/08 17:29:27 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform  
ble  
Spark context Web UI available at http://DESKTOP-JUFG59F:4040  
Spark context available as 'sc' (master = local[*], app id = local-1728401369663).  
Spark session available as 'spark'.  
Welcome to
```



```
      /--\     /\_/\_/_/_/_/\_/\_/\n    _/   \_\_//  _/  _/  _/  _/  _/\n   /_____\_\_/___/_/_/\_/_/\_/_/\_/\n          |__|       version 3.5.3  
         /__|\n        /__|
```

```
Using Scala version 2.12.18 (OpenJDK 64-Bit Server VM, Java 21.0.4)  
Type in expressions to have them evaluated.  
Type :help for more information.
```

```
scala>
```

To access Spark's Web UI, open a **web browser** and navigate to the following URL:

http://localhost:4040/

You can replace `localhost` with the name or IP address of your system. The Spark Web UI provides details about running jobs, stages, and resource usage.

Executors

[Show Additional Metrics](#)

Summary

	RDD Blocks	Storage Memory	Disk Used	Cores	Active Tasks	Failed Tasks	Complete Tasks	Total Tasks	Task Time (GC Time)	Input	Shuffle Read	Shuffle Write	Excluded
Active(1)	0	0.0 B / 434.4 MiB	0.0 B	4	0	0	0	0	1.2 min (0.3 s)	0.0 B	0.0 B	0.0 B	0
Dead(0)	0	0.0 B / 0.0 B	0.0 B	0	0	0	0	0	0.0 ms (0.0 ms)	0.0 B	0.0 B	0.0 B	0
Total(1)	0	0.0 B / 434.4 MiB	0.0 B	4	0	0	0	0	1.2 min (0.3 s)	0.0 B	0.0 B	0.0 B	0

Executors

Show entries Search:

Executor ID	Address	Status	RDD Blocks	Storage Memory	Disk Used	Cores	Active Tasks	Failed Tasks	Complete Tasks	Total Tasks	Task Time (GC Time)	Input	Shuffle Read	Shuffle Write	Thread Dump	Heap Histogram	Add Time	Remove Time
driver	DESKTOP-JUFGS9F:A9325	Active	0	0.0 B / 434.4 MiB	0.0 B	4	0	0	0	0	1.2 min (0.3 s)	0.0 B	0.0 B	0.0 B	Thread Dump	Heap Histogram	2024-10-08 17:34:43	-

Showing 1 to 1 of 1 entries

Previous **1** Next

Use the following command to exit Spark and close the Scala shell:

:quit

Alternatively, press **CTRL+D** while using the command prompt. Enter the following command to run Spark using Python:

pyspark

This launches the Spark shell with a Python interface.

```
C:\WINDOWS\system32>pyspark
Python 3.9.13 (tags/v3.9.13:6de2ca5, May 17 2022, 16:36:42) [MSC v.1929 64 bit (AMD64)]
Type "help", "copyright", "credits" or "license()" for more information.
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel)
24/10/09 17:11:55 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform where applicable
Welcome to

      /_/_/_/_/_\
     /  V  _  V  \
    /___/ . _\ \ , // /_/_/_\   version 3.5.3
   /__/_/

Using Python version 3.9.13 (tags/v3.9.13:6de2ca5, May 17 2022 16:36:42)
Spark context Web UI available at http://DESKTOP-JUFG59F:4040
Spark context available as 'sc' (master = local[*], app id = local-1728486717729).
SparkSession available as 'spark'.
>>> █
```

To exit **pyspark**, type:

```
quit()
```

Test Spark

To test the Spark installation, use the Scala interface to read and manipulate a file. In this example, the name of the file is *pnaptest.txt*. Open Command Prompt and navigate to the folder with the file you want to use:

1. Launch the Spark shell:

```
spark-shell
```

2. Load the file in the Spark context by defining a variable and specifying the file name (including its extension):

```
val x = sc.textFile("pnaptest.txt")
```

The command loads the file into a **Resilient Distributed Dataset (RDD)**, which allows you to perform actions and transformations on the data.

3. Enter the following command to perform an action on the RDD and view the contents of the file:

```
x.take(11).foreach(println)
```

```
scala> val x = sc.textFile("pnaptest.txt")
x: org.apache.spark.rdd.RDD[String] = pnaptest.txt MapPartitionsRDD[1] at textFile at <console>:23

scala> x.take(11).foreach(println)
Varying data structures
Structured data is characterized by a high degree of organization and is typically
the kind of data you see in relational databases or spreadsheets. Because
of its defined structure, it maps easily to one of the standard data types (or
user-defined types that are based on those standard types). It can be searched
using standard search algorithms and manipulated in well-defined ways.
Semistructured data (such as what you might see in log files) is a bit more difficult
to understand than structured data. Normally, this kind of data is stored in
the form of text files, where there is some degree of order ? for example, tabdelimited
files, where columns are separated by a tab character. So instead of
being able to issue a database query for a certain column and knowing exactly

scala>
```

Spark displays the first **11** lines of the file. You can adjust the number of lines by changing the number in the `take()` method.

4. Use the `map` transformation to create a new RDD based on the existing one. For example, reverse each line of the file using the following command:

```
val y = x.map(_.reverse)
```

This creates a new RDD (`y`), where each original RDD (`x`) line is reversed.

5. Enter the following command to display the first 11 lines of the transformed RDD:

```
y.take(11).foreach(println)
```

```
scala> val y = x.map(_.reverse)
y: org.apache.spark.rdd.RDD[String] = MapPartitionsRDD[2] at map at <console>:23

scala> y.take(11).foreach(println)
serutcurts atad gniyraV
yllacipyT si dna noitazinagro fo eerged hgiH a yb deziretcarahc si atad derutcurtS
esuaceB .steehsdaerps ro sesabatad lanoitaler ni ees uoy atad fo dnik eht
ro( sepyt atad dradnats eht fo eno ot ylisae spam ti ,erutcurts denifed sti fo
dehcræes eb nac tI .)sepyt dradnats esoht no desab era taht sepyt denifed-resu
.syaw denifed-llew ni detalupinam dna smhtirogla hcraes dradnats gnisu
tluciffid erom tib a si )selif gol ni ees thgim uoy tahw sa hcus( atad derutcurtsimeS
ni derots si atad fo dnik siht ,yllamroN .atad derutcurts naht dnatsrednu ot
detimiledbat ,elpmaxe rof ? redro fo eerged emos si ereht erehw ,selif txet fo mrof eht
fo daetsni oS .retcarahc bat a yb detarapes era snmuloc erehw ,selif
yltcaxe gniwonk dna nmuloc niatrec a rof yreuq esabatad a eussi ot elba gnieb

scala> _
```

The output prints 11 lines of the `pnaptest.txt` file, with each line reversed.

When ready, enter the `:quit` command or press **CTRL+D** to exit the Spark shell.

Conclusion

You now have a working installation of Apache Spark on Windows 10 or 11. This local setup is ideal for running basic tests and getting familiar with Spark's core features.

Read about [Spark DataFrames](#) and [Spark Streaming](#) to explore more advanced functionalities.

Was this article helpful?

Yes

No

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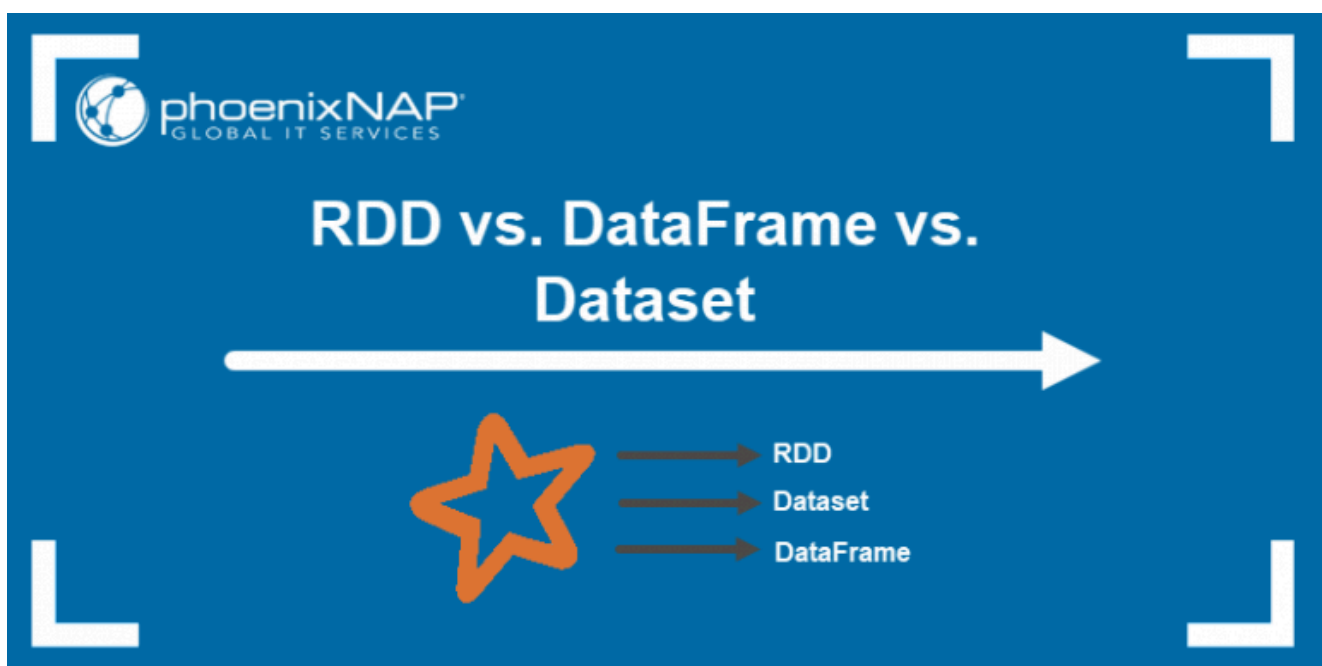
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2. Test Spark

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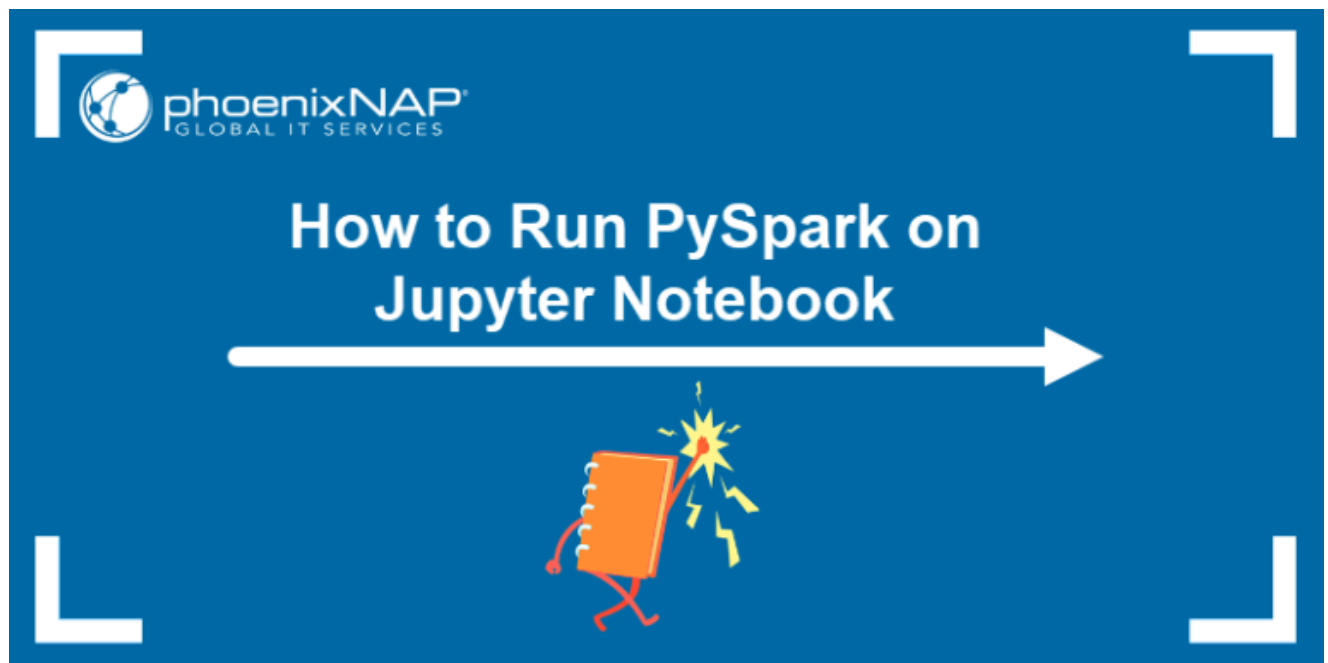
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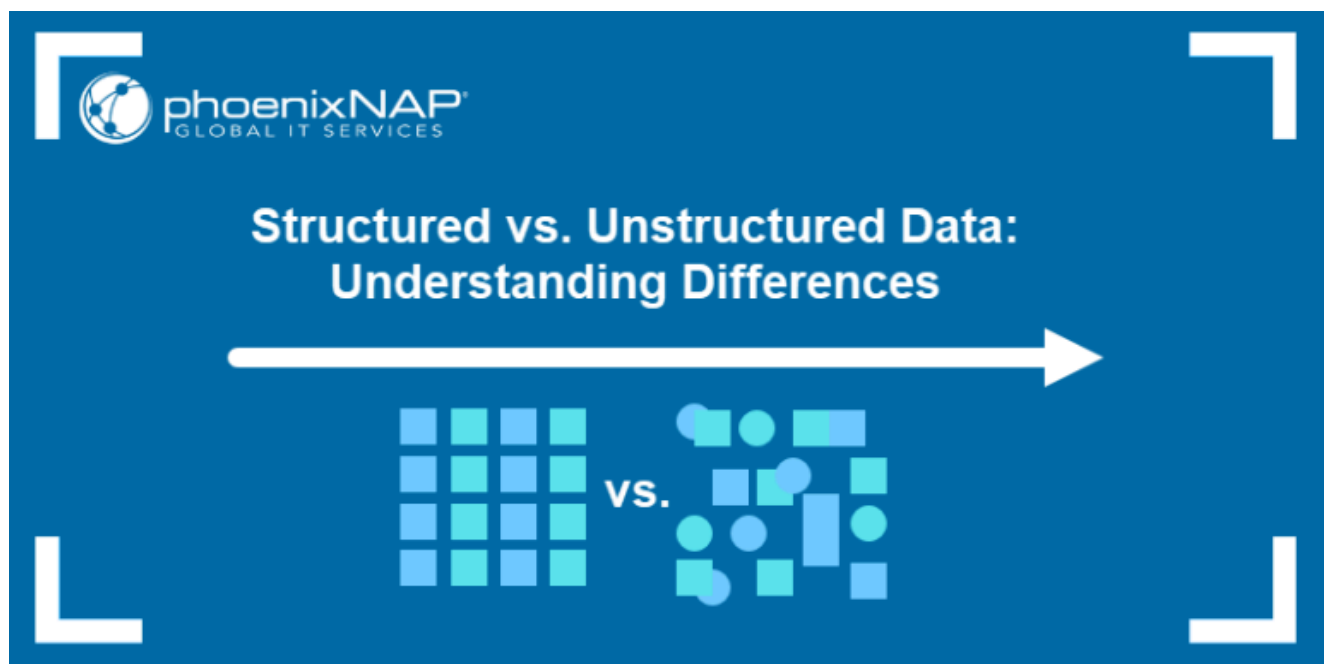
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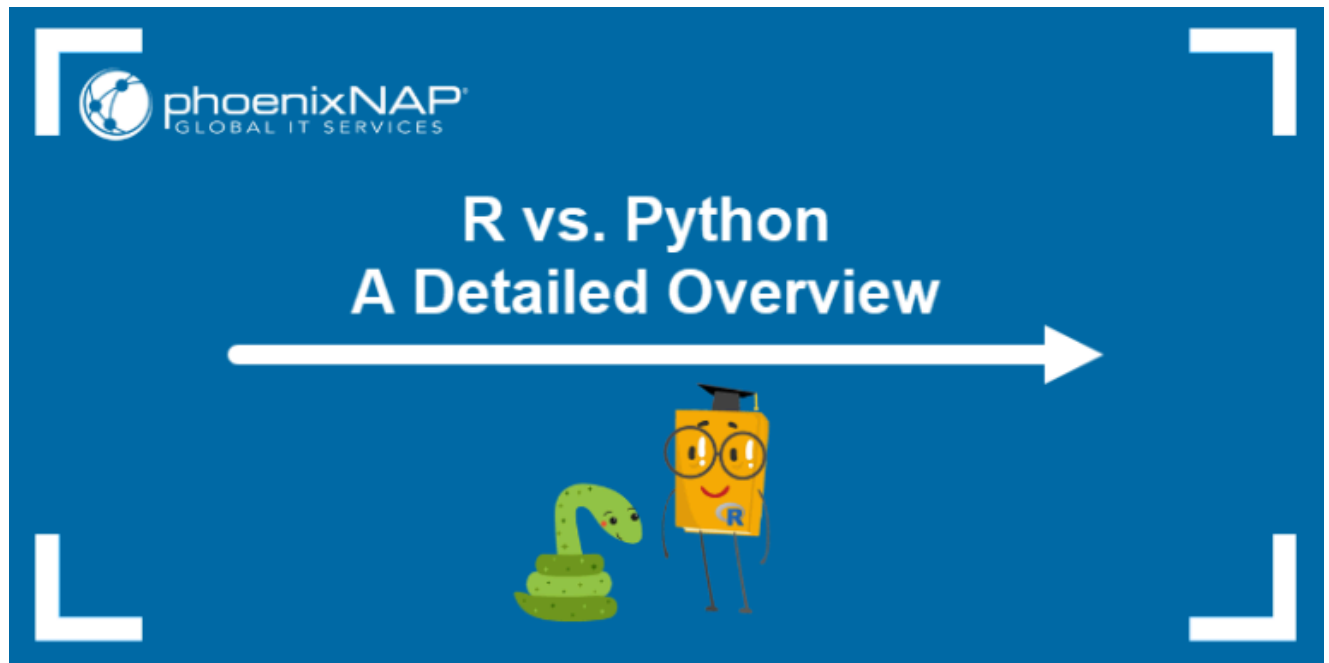
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